Generation of deep-sub-cycle optical pulses via inverse Compton scattering in a mid-infrared laser field

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ABSTRACT

Sub-cycle optical pulses with extremely high temporal gradients represent a forefront direction in ultrafast science. Although such pulses can provide ultrahigh temporal resolution for probing ultrafast dynamics, the ultimate half-cycle limit of pulse duration (i.e., pulse width smaller than half an optical cycle) has remained unbroken. To address this challenge, we recently proposed a scheme to generate deep-sub-cycle optical pulses via inverse Compton scattering (ICS) between a relativistic electron bunch and a deep-subwavelength confined terahertz (THz) optical field. Here, we extend this concept by replacing the THz field with a mid-infrared deep-subwavelength confined optical field, which can significantly enhance the energy of the emitted photons with similar electron-bunch incidence. For instance, when a 2-MeV, 50-as electron bunch interacts with a mid-infrared confined field with a peak wavelength of 4.4 μm and a pulse duration of 100 fs, the ICS process can generate a 0.26-cycle, 99-as extreme-ultraviolet (EUV) unipolar pulse with a peak frequency of 2.6 PHz. A trade-off between the temporal confinement and single-pulse photon number is also discussed.
Keywords: Deep-sub-cycle, inverse Compton scattering, relativistic electron bunch, mid-infrared laser field

1. INTRODUCTION

Time and space are the two most fundamental physical scales for studying the world. Since Einstein introduced the relativistic view of spacetime, these two dimensions have often been treated as a dual pair of physical concepts. In a manner analogous to achieving higher spatial resolution by reducing the wavelength of light[1,2], the pursuit of higher temporal resolution has focused on reducing the pulse duration— either by increasing the photon energy (thus reducing the optical cycle) or by decreasing the number of optical cycles (thus compressing the pulse envelope). Nevertheless, the advancement of current technologies—such as optical parametric chirped pulse amplification (OPCPA)[3,4], optical arbitrary waveform generation (OAWG)[5,6], and relativistic electron sheet (RES)[7,8]—has not yet reached the half-cycle limit on the temporal scale.

Recently, we proposed a promising approach to converting deep-sub-wavelength spatial confinement into deep-sub-cycle temporal confinement via inverse Compton scattering between a relativistic electron bunch and a deep-sub-wavelength confined optical field[9]. This approach achieves an ultrahigh temporal gradient in the near-field regime, thereby realizing higher temporal resolution. For instance, ICS between a 3-MeV 1-fs electron bunch and a 0.4-THz 1-ps confined optical field can yield a 0.1-cycle pulse with a duration of 3.6 fs and a peak frequency of 26 THz.

Here, we propose to generate deep-sub-cycle optical pulses via inverse Compton scattering in a mid-infrared confined optical field, to substantially enhancing the ICS photon energy with similar electron-bunch incidence. Based on the inverse Compton scattering of a relativistic electron bunch from a deep-sub-wavelength confined optical field. Our calculations indicate that inverse Compton scattering between a 2-MeV 50-as electron bunch and a 4.4-μm 100-fs deep-sub-wavelength confined driving field can generate a 0.26-cycle 99-as extreme ultraviolet (EUV) unipolar pulse, with a peak frequency of 2.6 PHz (i.e., ~115 nm in wavelength, within ultrashort extreme ultraviolet band).

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2. DEEP-SUB-CYCLE OPTICAL PULSES OBTAINED WITH MID-INFRARED LASER

2.1 Basic formulas

According to our previous work[9], the photons radiated from ICS between a driving laser pulse (wavelength λ_0, corresponding to a frequency ν_0; pulse duration τ_0; and single-pulse energy E_{p0}) and an electron bunch (energy E_e, corresponding to a velocity $V_e=\beta c$ where c is the speed of light in vacuum; and duration τ_e) can be described by the following equation:

$$n(z) = N_0 N_e \sigma_T \frac{S(z)}{P_{\text{mode}}} \frac{\delta_t}{\tau_0} \tag{1}$$

where N_e is the number of incident electrons and N_0 is the total number of photons in the driving field, approximated as $N_0 \approx E_{p0}/(h\nu_0)$, where h is the Planck constant. The fundamental interaction strength is determined by the Thomson scattering cross-section, σ_T, which has a value of 6.65×10^{-29} m^2. The term $S(z)$ represents the magnitude of the Poynting vector as a function of position z, while P_{mode} is the mode power of the optical field, defined as the integral of the Poynting vector over the mode area at the instant of the electron's transit. The temporal parameters are the electron transit time, δ_t, and the driving pulse duration, τ_0, which is assumed to be longer than the transit time ($\tau_0>\delta_t$).

In the spatial far field, the transverse scattered electric field $E_y(t)$ can be obtained as

$$E_y(t,\theta) = 2\sqrt{\frac{\mu_0}{\varepsilon_0}\cdot I(t,\theta)}\, e^{i\cdot 2\pi\nu_0 \frac{t}{1-\beta cos\theta}} \tag{2}$$

where ε_0 and μ_0 are the vacuum permittivity and the magnetic permeability, respectively and $I(t,\theta) = n(z)\,\chi^2(1+2\chi(\chi-1))$ is the radiated intensity of the ICS pulse at time t and scattered angle θ with $\chi=1/(1+\theta^2/(1+\beta^2))$. Integration over θ yields the total intensity $I(t)$ and electric field $E_y(t)$, from which the spectrum can be obtained via Fourier transform.

2.2 Results and Discussion

As depicted in Fig. 1a, a mid-infrared (mid-IR) laser pulse—with a wavelength of 4.4 µm, a duration of 100 fs, and a single-pulse energy of 0.4 µJ—is confined within a slit-coupled structure (SCS) by coupling two sapphire prisms. This arrangement generates a deep-sub-wavelength confined optical field. Assuming a slit width of 200 nm and a prism diagonal diameter of 1.2 µm, the central peak of the field exhibits a spatial Full Width at Half Maximum ($FWHM_S$) of approximately 808 nm along the z-direction, corresponding to a confinement of ~$0.18\lambda_0$. Meanwhile, we consider an electron bunch with a kinetic energy of 2 MeV (with a velocity $V_e\approx0.98$c) that traverses this confined field from left to right along the central axis of symmetry (i.e., the z-direction). The electron bunch[10-12] is assumed to have a Gaussian temporal profile with a duration of 50 as, a total charge of 5 pC, and an elliptical cross-section with short and long axes of 200 nm and 1 µm, respectively.

Based on the aforementioned parameters, our calculation results are presented in Figure 1. The generated pulse (Fig. 1b) has a duration (i.e., temporal Full Width at Half Maximum abbreviated as $FWHM_T$) of only 99 as. Despite some temporal broadening from the 50 as electron bunch, this is significantly narrower than the pulse generated from a conventional diffraction-limited driving field (red dotted line). The pulse's temporal profile, E_y(t), clearly preserves a unipolar waveform (Fig. 1c). Its corresponding frequency spectrum (Fig. 1d) is centered at a peak frequency of ν_p=2.6 PHz, which corresponds to an optical period of $T_c\approx0.38$ fs and a time confinement factor $\zeta=T_c/\tau_1\approx3.8$, confirming its sub-half-cycle feature (i.e., $\zeta>2$).

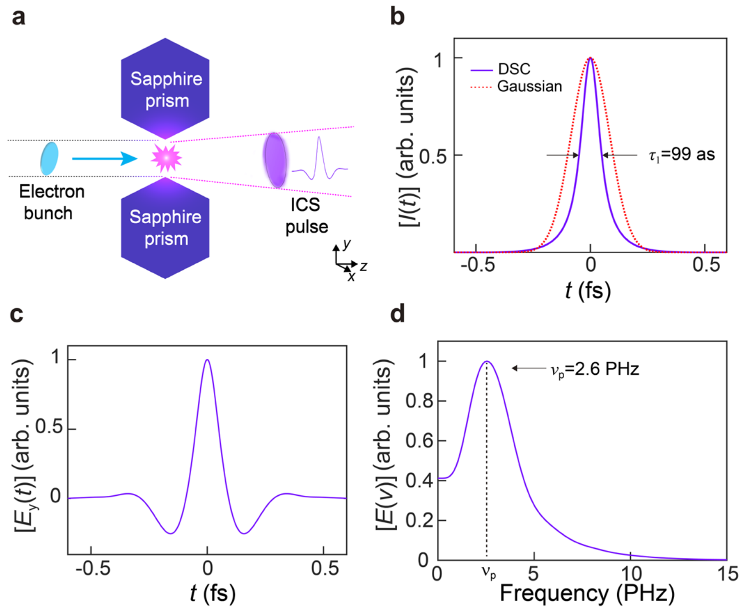

Figure. 1 | Generation of deep-sub-cycle attosecond pulses with attosecond electron bunch and mid-infrared optical driving field. a, Schematic diagram of the ICS system. A focused electron bunch with an elliptical cross-section is incident from the left side of a slit, collides with photons in the confined optical driving field that is generated along the slit between two coupled sapphire prisms. **b,** $I(t)$ of the ICS pulse, generated via ICS of a 50-as 2-MeV electron bunch in a pulsed driving field with a $\mathrm{FWHM_S}$ of 808 nm, a wavelength of 4.4 μm and a pulse width of 100 fs, yielding a pulse width (τ_I) of 99 as. For comparison, $I(t)$ of an ICS pulse generated in a 4.4-μm-wavelength Gaussian-profile optical driving field is also provided (red dotted line). DSC: deep-sub-cycle pulse, Gaussian: Gaussian-profile pulse. **c,** $E_y(t)$ of the DSC pulse in (**b**). **d,** Fourier spectrum $E(\nu)$ of the pulse in (**c**), giving a peak frequency of ~2.6 PHz.

The time confinement factor, ζ, can be changed over a wide range by changing the wavelength (λ_0) and spatial confinement ($\mathrm{FWHM_S}$) of the driving optical field. Figure 2a illustrates this dependence for the same 2-MeV electron bunch, showing that increasing λ_0 from 1 to 5 μm increases ζ from approximately 1 to over 5; similarly, for a given wavelength, a stronger confinement (smaller $\mathrm{FWHM_S}$) also results in a larger ζ. However, this enhanced deep-sub-cycle character comes at the cost of photon flux, as shown in Fig. 2b. A clear trade-off exists between the deep-sub-cycle parameter and the scattered photon number (N_{SP}). For instance, as ζ increases from 4.0 to 4.2, N_{SP} drops sharply from 270 to 130. Conversely, by reducing the confinement to decrease ζ to 3.2 (a value still well within the deep-sub-cycle regime), the photon number increases to 1260. This demonstrates a trade-off between the degree of temporal confinement and the overall photon yield of the generated pulse.

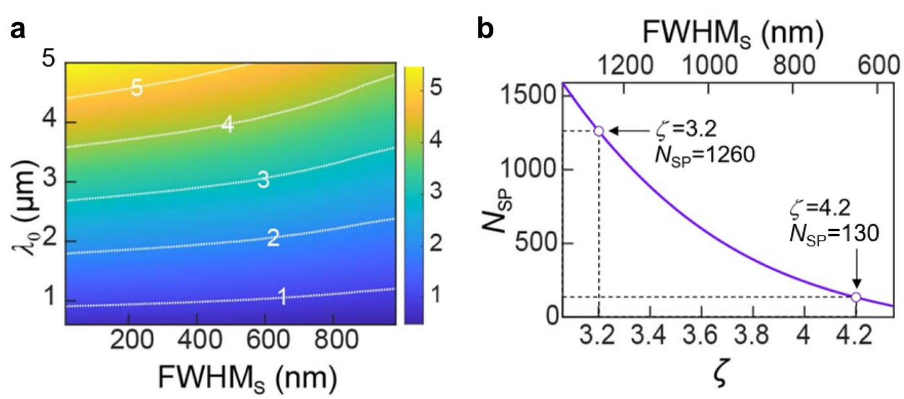

Figure. 2 | Trade-off between the temporal confinement and number of scattered photons. a, Dependence of ζ on $\mathrm{FWHM_S}$ and λ_0 of the optical driving field generated in two coupled sapphire prisms in Fig. 1, with a 50-as 2-MeV electron bunch. **b,** Dependence of N_{SP} on ζ and FWHM$_S$, with an optical driving field of 4.4-μm wavelength and a 2×10^{18} W/m^2 peak power density, and a 50-as 2-MeV 5-pC electron bunch.

3. CONCLUSION

In summary, we have proposed to generate deep-sub-cycle optical pulses via inverse Compton scattering in a mid-infrared laser field. Our calculations show that, based on inverse Compton scattering between a 2-MeV 50-as electron bunch and a deep-sub-wavelength confined 4.4-μm 100-fs mid-infrared laser field, it is possible to generate a 0.26-cycle, 99-as extreme-ultraviolet (EUV) unipolar pulse with a peak frequency of 2.6 PHz . Compared to the previously used THz driving field, here the mid-infrared field can provide significantly increased photon energy, and thus yield much higher-energy scattered photons with similar electron-bunch incidence.

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